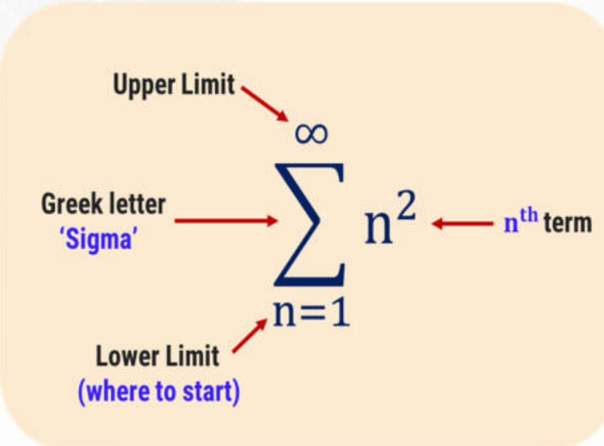
Mathematics-I
GTU#(3110014)

Unit-2

Sequence and Series



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Method:14 ➡ Leibnitz's Test

Alternating Series

- A series with alternate positive and negative terms is known as an alternating series.

Ex: $1 - 2 + 3 - 4 + \cdots = \sum_{n=1}^{\infty} (-1)^{n+1} \cdot n$

Method:14 ➔ Leibnitz's Test

Leibnitz's Test

- An alternating series $\sum a_n = \sum (-1)^{n+1} u_n = u_1 - u_2 + u_3 - \cdots$

  is **convergent** if

  1) $u_1 \geq u_2 \geq u_3 \geq u_4 \geq \cdots$

  2) $\lim\limits_{n \to \infty} u_n = 0$

- In above test, if $\lim\limits_{n \to \infty} u_n \neq 0$ then $\sum a_n$ is oscillating series.

Method:14 → Leibnitz's Test

Example:1 Check the convergence of $1-\frac{1}{2\sqrt{2}}+\frac{1}{3\sqrt{3}}-\frac{1}{4\sqrt{4}}+\cdots$.

Solution: Here, $1-\frac{1}{2\sqrt{2}}+\frac{1}{3\sqrt{3}}-\frac{1}{4\sqrt{4}}+\cdots=\sum_{n=1}^{\infty}(-1)^{n+1}\cdot\frac{1}{n\sqrt{n}}$

$$\therefore u_n=\frac{1}{n\sqrt{n}}$$

Now, $u_1=1,\ u_2=\frac{1}{2\sqrt{2}},\ u_3=\frac{1}{3\sqrt{3}}$ and so on.

$$\therefore u_1\geq u_2\geq u_3\geq u_4\geq\cdots$$

Method:14 ⇝ Example:1(Continue)

$\therefore u_1 \geq u_2 \geq u_3 \geq u_4 \geq \cdots$

➡ Now, $\lim\limits_{n \to \infty} u_n = \lim\limits_{n \to \infty} \dfrac{1}{n\sqrt{n}}$

$$= \lim_{n \to \infty} \frac{1}{n^{\frac{3}{2}}}$$

$$= 0$$

$\because \lim\limits_{n \to \infty} \left(\dfrac{1}{n}\right) = 0$

➡ Thus, by **Leibnitz's test**, the given series is **convergent**.

Method:14 → Leibnitz's Test

Example:2 Check the convergence of $\frac{1}{1^2+1}-\frac{2}{2^2+1}+\frac{3}{3^2+1}-\cdots$.

Solution: Here, $\frac{1}{1^2+1}-\frac{2}{2^2+1}+\frac{3}{3^2+1}-\cdots=\sum_{n=1}^{\infty}(-1)^{n+1}\cdot\frac{n}{n^2+1}$

$$\therefore u_n=\frac{n}{n^2+1}$$

Now, $u_1=\frac{1}{2}$, $u_2=\frac{2}{5}$, $u_3=\frac{3}{10}$ and so on.

$$\therefore u_1 \geq u_2 \geq u_3 \geq \cdots$$

Method:14 ⇝ Example:2(Continue)

$\therefore u_1 \geq u_2 \geq u_3 \geq \cdots$

- Now, $\lim\limits_{n \to \infty} u_n = \lim\limits_{n \to \infty} \dfrac{n}{n^2 + 1}$

$$= \lim_{n \to \infty} \frac{n}{n^2\left(1 + \frac{1}{n^2}\right)}$$

$$= \lim_{n \to \infty} \frac{1}{n\left(1 + \frac{1}{n^2}\right)}$$

$$\because \lim_{n \to \infty} \left(\frac{1}{n}\right) = 0$$

$$= 0$$

- Thus, by **Leibnitz's test**, the given series is **convergent**.

Method:14 → Leibnitz's Test

Example:3 Test the convergence of the series $\frac{1}{1^2}-\frac{1}{2^2}+\frac{1}{3^2}-\frac{1}{4^2}+\cdots$.

Solution: Here, $\frac{1}{1^2}-\frac{1}{2^2}+\frac{1}{3^2}-\frac{1}{4^2}+\cdots=\sum_{n=1}^{\infty}(-1)^{n+1}\cdot\frac{1}{n^2}$

$$\therefore u_n=\frac{1}{n^2}$$

Now, $u_1=1, u_2=\frac{1}{4}, u_3=\frac{1}{9}, u_4=\frac{1}{16}$ and so on.

$$\therefore u_1 \geq u_2 \geq u_3 \geq u_4 \geq \cdots$$

Method:14 ⇝ Example:3(Continue)

$\therefore u_1 \geq u_2 \geq u_3 \geq u_4 \geq \cdots$

$\therefore \{u_n\}$ is decreasing.

- Now, $\lim\limits_{n \to \infty} u_n = \lim\limits_{n \to \infty} \dfrac{1}{n^2}$

$$= 0$$

$$\because \lim_{n \to \infty} \left(\frac{1}{n}\right) = 0$$

- Thus, by **Leibnitz's test**, the given series is **convergent**.

Method:14 → Leibnitz's Test

Example:4 Check the convergence of

$$\left(\frac{1}{2}+\frac{1}{\ln 2}\right)-\left(\frac{1}{2}+\frac{1}{\ln 3}\right)+\left(\frac{1}{2}+\frac{1}{\ln 4}\right)-\cdots.$$

Solution: Here, $\left(\frac{1}{2}+\frac{1}{\ln 2}\right)-\left(\frac{1}{2}+\frac{1}{\ln 3}\right)+\left(\frac{1}{2}+\frac{1}{\ln 4}\right)-\cdots$

$$=\sum_{n=1}^{\infty}(-1)^{n+1}\cdot\left(\frac{1}{2}+\frac{1}{\ln(n+1)}\right)$$

Method:14 ⇝ Example:4(Continue)

$$= \sum_{n=1}^{\infty} (-1)^{n+1} \cdot \left(\frac{1}{2} + \frac{1}{\ln(n+1)} \right)$$

$$\therefore u_n = \frac{1}{2} + \frac{1}{\ln(n+1)}$$

Now, $u_1 = \frac{1}{2} + \frac{1}{\ln(2)} \approx 3.822$

$$u_2 = \frac{1}{2} + \frac{1}{\ln(3)} \approx 2.595$$

$u_3 = \frac{1}{2} + \frac{1}{\ln(4)} \approx 2.160$ and so on.

$$(n+1) \leq (n+2)\,;\, n \in (0, \infty)$$

$$\Longrightarrow \log(n+1) \leq \log(n+2)$$

$$\Longrightarrow \frac{1}{\log(n+1)} \geq \frac{1}{\log(n+2)}$$

$$\Longrightarrow \frac{1}{2} + \frac{1}{\log(n+1)} \geq \frac{1}{2} + \frac{1}{\log(n+2)}$$

$$\Longrightarrow u_n \geq u_{n+1}$$

Method:14 ⇝ Example:4(Continue)

$$u_3 = \frac{1}{2} + \frac{1}{\ln(4)} \approx 2.160 \text{ and so on.}$$

$$\therefore u_1 \geq u_2 \geq u_3 \geq u_4 \geq \cdots$$

➡ Now, $\lim\limits_{n \to \infty} u_n = \lim\limits_{n \to \infty} \left(\frac{1}{2} + \frac{1}{\ln(n+1)} \right)$

$$= \frac{1}{2} + 0$$

$\because \lim\limits_{n \to \infty} \log n = 0$

$$= \frac{1}{2} \neq 0$$

➡ Thus, by **Leibnitz's test**, the given series is **oscillating**.

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Method:15 → Absolute Convergent Series

Absolute Convergent Series

- An alternating series $\sum a_n$ is said to be absolute convergent, if $\sum |a_n|$ is convergent.

Note An absolute convergent series is always convergent.

Method:15 ➡ Absolute Convergent Series

Example:1 Is series $1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\frac{1}{5}-\cdots$ convergent?

Does it converge absolutely?

Solution: 1) Here, $1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\frac{1}{5}-\cdots=\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}$

$$\therefore u_n=\frac{1}{n}$$

Now, $u_1=1, u_2=\frac{1}{2}, u_3=\frac{1}{3}, u_4=\frac{1}{4}$ and so on.

$$\therefore u_1 \geq u_2 \geq u_3 \geq u_4 \geq \cdots$$

Method:15 ⇝ Example:1(Continue)

$\therefore u_1 \geq u_2 \geq u_3 \geq u_4 \geq \cdots$

➡ Now, $\lim\limits_{n \to \infty} u_n = \lim\limits_{n \to \infty} \frac{1}{n}$

$$= 0$$

➡ Thus, by **Leibnitz's test**, the given series is **convergent**.

2) Here, $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$

$$\therefore a_n = \frac{(-1)^{n+1}}{n}$$

Method:15 ⇝ Example:1(Continue)

$$\therefore a_n = \frac{(-1)^{n+1}}{n}$$

- Let, $w_n = |a_n|$

$$\therefore w_n = \frac{1}{n}$$

Now, $\sum_{n=1}^{\infty} w_n = \sum_{n=1}^{\infty} \frac{1}{n}$ is divergent by **p-series test**, as $p = 1$.

- Thus, the given series does not converges absolutely.

Method:15 Absolute Convergent Series

Example:2 Determine the absolute convergence

$$1 - 2x + 3x^2 - 4x^3 + \cdots \quad (0 \leq x < 1)$$

Solution: Here, $1 - 2x + 3x^2 - 4x^3 + \cdots = \sum_{n=1}^{\infty} (-1)^{n+1} \cdot n \cdot x^{n-1}$

$$\therefore a_n = (-1)^{n+1} \cdot n \cdot x^{n-1}$$

Let, $w_n = |a_n|$

$$\therefore w_n = n \cdot x^{n-1} \Rightarrow w_{n+1} = (n+1) \cdot x^n$$

Method:15 → Example:2(Continue)

$$\therefore w_n = n \cdot x^{n-1} \implies w_{n+1} = (n+1) \cdot x^n$$

Now, $\lim\limits_{n\to\infty} \dfrac{w_n}{w_{n+1}} = \lim\limits_{n\to\infty} \dfrac{n \cdot x^{n-1}}{(n+1) \cdot x^n}$

$$= \lim_{n\to\infty} \frac{n \cdot x^n \cdot x^{-1}}{n \cdot \left(1 + \frac{1}{n}\right) \cdot x^n}$$

$$= \frac{1}{x}$$

Given that, $0 \le x < 1$

$$\therefore \frac{1}{x} > 1$$

$$\therefore \lim_{n\to\infty} \frac{w_n}{w_{n+1}} = \frac{1}{x} > 1$$

Thus, by **Ratio test**,

$\sum\limits_{n=1}^{\infty} w_n$ is convergent.

Hence, $\sum\limits_{n=1}^{\infty} a_n$ converges absolutely.

Method:15 Absolute Convergent Series

Example:3 Determine the absolute convergence of $(1)\ \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^p}\ (p > 1)$

Solution: Here, $a_n = \frac{(-1)^{n-1}}{n^p}$

Let, $w_n = |a_n|$

$$\therefore w_n = \frac{1}{n^p}$$

Now, $\sum_{n=1}^{\infty} w_n = \sum_{n=1}^{\infty} \frac{1}{n^p}$ and given that $p > 1$

Method:15 ⇝ Example:3(1)(Continue)

Now, $\sum_{n=1}^{\infty} w_n = \sum_{n=1}^{\infty} \frac{1}{n^p}$ and given that $p > 1$

So, by **p-series** test, $\sum_{n=1}^{\infty} w_n = \sum_{n=1}^{\infty} |a_n|$ is convergent.

➡ Thus, the given series converges absolutely.

Method:15 → Absolute Convergent Series

Example:3 Determine the absolute convergence of (2) $\sum_{n=1}^{\infty}(-1)^{n+1}\left(\frac{\sin n}{n^2}\right)$

Solution: Here, $a_n = (-1)^{n+1}\left(\frac{\sin n}{n^2}\right)$

→ Let, $w_n = |a_n|$

$$\therefore w_n = \frac{|\sin n|}{n^2}$$

We know that, $-1 \le \sin n \le 1$

$$\Rightarrow 0 \le |\sin n| \le 1$$

Method:15 ⇝ Example:3(2)(Continue)

$$\Rightarrow 0 \le |\sin n| \le 1$$

$$\Rightarrow 0 \le \frac{|\sin n|}{n^2} \le \frac{1}{n^2}$$

Since, $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent by **p-series test**, as $p = 2 > 1$

$\Rightarrow \sum_{n=1}^{\infty} \frac{|\sin n|}{n^2}$ is convergent by **direct comparison test**.

$\Rightarrow \sum_{n=1}^{\infty} w_n = \sum_{n=1}^{\infty} |a_n|$ is convergent.

Method:15 ⇝ Example:3(2)(Continue)

$$\Rightarrow \sum_{n=1}^{\infty} w_n = \sum_{n=1}^{\infty} |a_n| \text{ is convergent.}$$

Thus, $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} (-1)^{n+1}\left(\frac{\sin n}{n^2}\right)$ is absolutely convergent.

Method:16 → Conditionally Convergent Series

Method:16 → Conditionally Convergent Series

Conditionally Convergent Series

- An alternating series $\sum a_n$ is said to be conditionally convergent, if

  1) $\sum a_n$ is **convergent**.

  2) $\sum |a_n|$ is **divergent**.

Note Infinite series can not be absolute and conditionally convergent together.

Method:16 ➔ Conditionally Convergent Series

Example:1 Check the absolute and conditionally convergence of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt[3]{n}}$.

Solution: Here, $a_n = \frac{(-1)^{n+1}}{\sqrt[3]{n}}$

➔ Let, $u_n = \frac{1}{\sqrt[3]{n}}$

$$\therefore u_n = \frac{1}{n^{\frac{1}{3}}}$$

1) $\sum a_n$ is **convergent**.

2) $\sum |a_n|$ is **divergent**.

Method:16 ⇝ Example:1(Continue)

$$\therefore u_n = \frac{1}{n^{\frac{1}{3}}}$$

Now, $u_1 = 1,\ u_2 = \frac{1}{2^{\frac{1}{3}}},\ u_3 = \frac{1}{3^{\frac{1}{3}}}$ and so on.

$$\therefore u_1 \geq u_2 \geq u_3 \geq \cdots$$

➡ Now, $\lim\limits_{n \to \infty} u_n = \lim\limits_{n \to \infty} \frac{1}{n^{\frac{1}{3}}}$

$$= 0$$

➡ Thus, by **Leibnitz's test**, $\sum_{n=1}^{\infty} a_n$ is convergent.

- Thus, by **Leibnitz's test**, $\sum_{n=1}^{\infty} a_n$ is convergent.
- Now, $\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} \left| \frac{(-1)^{n+1}}{\sqrt[3]{n}} \right|$

$$= \sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n}}$$

$= \sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{3}}}$ is divergent by **p-series test**, as $p = \frac{1}{3} < 1.$

1) $\sum a_n$ is **convergent**.

2) $\sum |a_n|$ is **divergent**.

Method:16 ⇝ Example:1(Continue)

$$= \sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{3}}}$$ is divergent by **p-series test**, as $p = \frac{1}{3} < 1.$

Thus, $\sum_{n=1}^{\infty} a_n$ is convergent and $\sum_{n=1}^{\infty} |a_n|$ is divergent.

Hence, the given series is **conditionally convergent**.

Method:16 → Conditionally Convergent Series

Example:2 Check the convergence of $-\frac{1}{1^p}+\frac{1}{2^p}-\frac{1}{3^p}+\frac{1}{4^p}-\cdots;\ p>0$

Solution: Here $-\frac{1}{1^p}+\frac{1}{2^p}-\frac{1}{3^p}+\frac{1}{4^p}-\cdots=\sum_{n=1}^{\infty}(-1)^n\frac{1}{n^p}$.

Here, $a_n=(-1)^n\frac{1}{n^p}$

→ Let, $u_n=\frac{1}{n^p}$

Now, $u_1=\frac{1}{n^p}, u_2=\frac{1}{2^p}, u_3=\frac{1}{3^p}$ and so on.

1) $\sum a_n$ is **convergent**.

2) $\sum |a_n|$ is **divergent**.

Method:16 ⇝ Example:2(Continue)

Now, $u_1 = \frac{1}{n^p}, u_2 = \frac{1}{2^p}, u_3 = \frac{1}{3^p}$ and so on.

$\therefore u_1 \geq u_2 \geq u_3 \geq \cdots (\because p > 0)$

- Now, $\lim\limits_{n \to \infty} u_n = \lim\limits_{n \to \infty} \frac{1}{n^p}$

  $= 0$ as $p > 0$

- Thus, by **Leibnitz's test**, $\sum\limits_{n=1}^{\infty} a_n$ is convergent.

Method:16 ⇝ Example:2(Continue)

- Thus, by **Leibnitz's test**, $\sum_{n=1}^{\infty} a_n$ is convergent.

- Now, $\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} \left|\frac{(-1)^n}{n^p}\right|$

$$= \sum_{n=1}^{\infty} \frac{1}{n^p}$$

Given that, $\mathbf{p > 0}$.

Method:16 ⇝ Example:2(Continue)

Given that, $p > 0$.

Case:1 If $p > 1$

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$ is convergent by **p-series test**, as $p > 1$.

The given series is **absolutely convergent** for $p > 1$ and is not conditionally convergent when $p > 1$.

Case:2 If $0 < p \leq 1$, $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is divergent by **p-series test**, as $0 < p \leq 1$.

Method:16 ⇝ Example:2(Continue)

Case:2 If $0 < p \le 1$ $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is divergent by **p-series test**, as $0 < p \le 1$.

➡ Thus, the given series is conditionally convergent for $0 < p \le 1$.

Conclusion The given series is **absolutely convergent** for $p > 1$ and **conditionally convergent** for $0 < p \le 1$.

Method:17 → Power Series

Power Series

- An infinite series of the form

$$\sum_{n=0}^{\infty} a_n (x-c)^n = a_0 + a_1(x-c) + a_2(x-c)^2 + a_3(x-c)^3 + \cdots$$

is known as power series, where a_n is the co-efficient of the n^{th} term, c is a constant and x varies around c.

- The series is also known as power series about $\mathbf{x = c}$.

Power Series

$$\sum_{n=0}^{\infty} a_n (x-c)^n = a_0 + a_1(x-c) + a_2(x-c)^2 + a_3(x-c)^3 + \cdots$$

- If $\mathbf{c = 0}$ then, it is called power series in terms of $\mathbf{x}$.

$$\sum_{n=0}^{\infty} a_n (x)^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots$$

Radius & Interval of Convergence for Power Series

- Let $\sum_{n=0}^{\infty} a_n (x-c)^n$ be a power series about $x = c$ and $|x-c| < R$

  where, $R = \lim_{n \to \infty} \left| \frac{a_n}{a_{n+1}} \right|$

  Then, R called radius of convergence and $(c-R,\ c+R)$ is called interval of convergence.

Radius & Interval of Convergence for Power Series

- An interval in which power series converges is called the interval of convergence and the **half length** of the interval of convergence is called the radius of convergence.
- Radius of the convergence for a power series can be obtained from either of the formulae as follows:

$$R = \lim_{n \to \infty} \left| \frac{a_n}{a_{n+1}} \right| \quad \textbf{OR} \quad R = \lim_{n \to \infty} \frac{1}{(a_n)^{\frac{1}{n}}}$$

Convergence of Power Series

- Let $\sum_{n=0}^{\infty} a_n (x - c)^n$ be a power series. Then exactly one of the following conditions hold:

  1) The series always converges at $x = 0$.
  2) The series always converges for all x.
  3) There exists a positive number R such that series converges for $|x - c| < R$ and diverges for $|x - c| > R$.

NOTES

- If a power series is convergent for all values of x, then interval of convergence will be $(-\infty, \infty)$ and the radius of convergence will be ∞.
- For any x, outside the interval $(x - R, x + R)$, a power series diverges.

Method:17 → Power Series

Example:1 Find the radius of convergence and interval of convergence of

$$1-\frac{1}{2}(x-2)+\frac{1}{2^2}(x-2)^2+\cdots+\left(-\frac{1}{2}\right)^n(x-2)^n+\cdots.$$

Solution: Here $-1-\frac{1}{2}(x-2)+\frac{1}{2^2}(x-2)^2+\cdots+\left(-\frac{1}{2}\right)^n(x-2)^n+\cdots$

$$=\sum_{n=1}^{\infty}\left(-\frac{1}{2}\right)^n(x-2)^n$$

Here, $a_n=\left(-\frac{1}{2}\right)^n(x-2)^n$

Method:17 ⇝ Example:1(Continue)

Here, $a_n = \left(-\frac{1}{2}\right)^n (x-2)^n$

$$\Rightarrow a_{n+1} = \left(-\frac{1}{2}\right)^{n+1} (x-2)^{n+1}$$

Now, $\lim_{n\to\infty}\left|\frac{a_n}{a_{n+1}}\right| = \lim_{n\to\infty}\left|\frac{\left(-\frac{1}{2}\right)^n (x-2)^n}{\left(-\frac{1}{2}\right)^{n+1} (x-2)^{n+1}}\right|$

Method:17 ⇝ Example:1(Continue)

Now, $\lim\limits_{n \to \infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim\limits_{n \to \infty} \left| \frac{\left(-\frac{1}{2}\right)^n (x-2)^n}{\left(-\frac{1}{2}\right)^{n+1} (x-2)^{n+1}} \right|$

$$= \lim_{n \to \infty} \left| \frac{\left(-\frac{1}{2}\right)^n (x-2)^n}{\left(-\frac{1}{2}\right)^n \left(-\frac{1}{2}\right) (x-2)^n (x-2)} \right|$$

$$= \lim_{n \to \infty} \left| \frac{1}{\left(-\frac{1}{2}\right)(x-2)} \right|$$

Method:17 ⇝ Example:1(Continue)

$$= \lim_{n \to \infty} \left| \frac{1}{\left(-\frac{1}{2}\right)(x-2)} \right|$$

$$= \left| \frac{-2}{(x-2)} \right|$$

$$= \frac{2}{|x-2|}$$

Now, by **ratio test** the given series is convergent, if $\dfrac{2}{|x-2|} > 1$

$$\Rightarrow 2 > |x-2|$$

$$\Rightarrow 2 > |x - 2|$$

$$\Rightarrow |x - 2| < 2 = \mathbf{R}$$

$$\Rightarrow -2 < x - 2 < 2$$

$$\Rightarrow 0 < x < 4$$

- At $\mathbf{x = 0}$

$$a_n = \left(-\frac{1}{2}\right)^n (-2)^n$$

$$\Rightarrow a_n = \frac{(-1)^n}{2^n}(-1)^n(2)^n$$

$$\Rightarrow a_n = 1$$

Now, $\sum_{n=1}^{\infty} 1$ is divergent.

$$a_n = \left(-\frac{1}{2}\right)^n (x-2)^n$$

- At $\mathbf{x = 4}$

$$a_n = \left(-\frac{1}{2}\right)^n (2)^n \Rightarrow a_n = (-1)^n$$

Method:17 ⇝ Example:1(Continue)

$\Longrightarrow a_n = (-1)^n$

Now, $\sum_{n=1}^{\infty} (-1)^n$ is not convergent.

➡ Thus, the interval of convergence is $\mathbf{0 < x < 4}$. i.e. $\mathbf{(0, 4)}$

and the radius of convergence $\mathbf{R = 2}$.

Method:17 → Power Series

Example:2 Find the radius of convergence and interval of convergence of

$$\sum_{n=0}^{\infty} n! \cdot x^n$$

Solution: Here, $a_n = n!\,x^n \Longrightarrow a_{n+1} = (n+1)!\,x^{n+1}$

→ Now, $\lim_{n \to \infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim_{n \to \infty} \left| \frac{n!\,x^n}{(n+1)!\,x^{n+1}} \right|$

$$= \lim_{n \to \infty} \left| \frac{1}{n\left(1 + \frac{1}{n}\right)x} \right|$$

Method:17 ⇝ Example:2(Continue)

$$= \lim_{n \to \infty} \left| \frac{1}{n\left(1 + \frac{1}{n}\right)x} \right|$$

$$= 0$$

Thus, the series **converges** only when $\mathbf{x = 0}$ and **diverges** for all $\mathbf{x \neq 0}$.

Hence, radius of convergence $\mathbf{R = 0}$ and there is no interval of convergence.

Method:17 → Power Series

Example:3 Find the radius of convergence and interval of convergence of

$$\sum_{n=0}^{\infty} \frac{x^{2n-1}}{(2n-1)!}$$

Solution: Here, $a_n = \dfrac{x^{2n-1}}{(2n-1)!} \Longrightarrow a_{n+1} = \dfrac{x^{2n+1}}{(2n+1)!}$

→ Now, $\lim\limits_{n\to\infty}\left|\dfrac{a_n}{a_{n+1}}\right| = \lim\limits_{n\to\infty}\left|\dfrac{x^{2n-1}(2n+1)!}{(2n-1)!\,x^{2n+1}}\right|$

Method:17 ⇝ Example:3(Continue)

➡ Now, $\lim_{n\to\infty}\left|\frac{a_n}{a_{n+1}}\right| = \lim_{n\to\infty}\left|\frac{x^{2n-1}(2n+1)!}{(2n-1)!\,x^{2n+1}}\right|$

$$= \lim_{n\to\infty}\left|\frac{x^{2n}x^{-1}(2n+1)(2n)(2n-1)!}{(2n-1)!\,x^{2n}\cdot x}\right|$$

$$= \lim_{n\to\infty}\left|\frac{(2n+1)(2n)}{x^2}\right|$$

$$= \lim_{n\to\infty}\left|\frac{4n^2\left(1+\frac{1}{2n}\right)}{x^2}\right|$$

Method:17 ⇝ Example:3(Continue)

$$= \lim_{n \to \infty} \left| \frac{4n^2\left(1 + \frac{1}{2n}\right)}{x^2} \right|$$

$$= \infty$$

Thus, radius of convergence **R**: ∞

and interval of convergence: $(-\infty, \infty)$

Method:17 → Power Series

Example:4 For the series $\sum_{n=1}^{\infty}(-1)^n \cdot \frac{(x+2)^n}{n}$ find the radius and interval of convergence. For what values of x does the series converges absolutely, conditionally ?

Solution: Here, $a_n = (-1)^n \frac{(x+2)^n}{n} \Rightarrow a_{n+1} = (-1)^{n+1} \frac{(x+2)^{n+1}}{n+1}$

→ Now, $\lim_{n \to \infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim_{n \to \infty} \left| \frac{(-1)^n (x+2)^n}{n} \cdot \frac{n+1}{(-1)^{n+1}(x+2)^{n+1}} \right|$

Method:17 ⇝ Example:4(Continue)

➡ Now, $\lim\limits_{n\to\infty}\left|\dfrac{a_n}{a_{n+1}}\right| = \lim\limits_{n\to\infty}\left|\dfrac{(-1)^n(x+2)^n}{n}\cdot\dfrac{n+1}{(-1)^{n+1}(x+2)^{n+1}}\right|$

$$= \lim_{n\to\infty}\left|\frac{(-1)^n(x+2)^n}{n}\cdot\frac{n+1}{(-1)^n(-1)(x+2)^n(x+2)}\right|$$

$$= \lim_{n\to\infty}\left|\frac{n\left(1+\frac{1}{n}\right)}{n(-1)(x+2)}\right|$$

$$= \frac{1}{|x+2|}$$

Method:17 ⇝ Example:4(Continue)

$$= \frac{1}{|x+2|}$$

Now, by **ratio test** the given series is convergent, if $\frac{1}{|x+2|} > 1$

$$\Rightarrow 1 > |x+2|$$

$$\Rightarrow |x+2| < 1 = \mathbf{R}$$

$$\Rightarrow -1 < x+2 < 1$$

$$\Rightarrow -3 < x < -1$$

Method:17 ⇝ Example:4(Continue)

$\Rightarrow -3 < x < -1$

$$a_n = (-1)^n \frac{(x+2)^n}{n}$$

- At $\mathbf{x = -3}$

$$a_n = \frac{(-1)^n(-1)^n}{n}$$

$$a_n = \frac{1}{n}$$

Now, $\sum_{n=0}^{\infty} \frac{1}{n}$ is divergent by **p-series test** as $p = 1$.

Method:17 ⇝ Example:4(Continue)

- At $\mathbf{x = -1}$

$$a_n = \frac{(-1)^n (1)^n}{n} = \frac{(-1)^n}{n}$$

➡ Let, $u_n = \frac{1}{n}$

Now, $u_1 = 1$, $u_2 = \frac{1}{2}$, $u_3 = \frac{1}{3}$ and so on.

$\therefore u_1 \geq u_2 \geq u_3 \geq \cdots$

➡ Now, $\lim_{n \to \infty} u_n = \lim_{n \to \infty} \frac{1}{n}$

Method:17 ⇝ Example:4(Continue)

- Now, $\lim_{n \to \infty} u_n = \lim_{n \to \infty} \frac{1}{n}$

$$= 0$$

- Thus, by **Leibnitz's test**, $\sum_{n=1}^{\infty} a_n$ is convergent for $x = -1$
- Thus, radius of convergence $\mathbf{R}$: $\mathbf{1}$ as $|\mathbf{x} + \mathbf{2}| < \mathbf{1}$

  and interval of convergence: $(\mathbf{-3}, \mathbf{-1}]$
- Now, $\sum_{n=1}^{\infty} a_n$ is convergent for $x = -1$

Now, $\sum_{n=1}^{\infty} a_n$ is convergent for $x = -1$

where, $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$

1) $\sum a_n$ is **convergent**.

2) $\sum |a_n|$ is **divergent**.

Now, $\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} \left|\frac{(-1)^n}{n}\right|$

$= \sum_{n=1}^{\infty} \frac{1}{n}$ is divergent by **p-series test**, as $\mathbf{p = 1}$.

Method:17 ⇝ Example:4(Continue)

$$= \sum_{n=1}^{\infty} \frac{1}{n}$$ is divergent by **p-series test**, as $p = 1$.

➡ Thus, the given series is conditionally convergent at $x = -1$.

Conclusion The given series is **absolutely convergent** in $(-3, -1)$ and **conditionally convergent** at $x = -1$.

Method:18 ➡ 1^{st} form of Taylor's Series

Taylor's Series: 1^{st} Form

- If $f(x)$ possesses derivatives of all order at a point 'a', then Taylor's series of given function $f(x)$ at a point 'a', is given by

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \cdots$$

Method:18 → 1^{st} Form of Taylor's Series

Example:1 Expand e^x in power of $(x-1)$ up to first four terms.

Solution: Here, $f(x) = e^x$

By Taylor's Series,

$$\mathbf{f(x)} = \mathbf{f(a)} + \frac{\mathbf{f'(a)}}{\mathbf{1!}}(\mathbf{x}-\mathbf{a}) + \frac{\mathbf{f''(a)}}{\mathbf{2!}}(\mathbf{x}-\mathbf{a})^2 + \frac{\mathbf{f'''(a)}}{\mathbf{3!}}(\mathbf{x}-\mathbf{a})^3 + \cdots$$

Now, putting $\mathbf{a} = 1$

$$\mathbf{f(x)} = \mathbf{f(1)} + \frac{\mathbf{f'(1)}}{\mathbf{1!}}(\mathbf{x}-\mathbf{1}) + \frac{\mathbf{f''(1)}}{\mathbf{2!}}(\mathbf{x}-\mathbf{1})^2 + \frac{\mathbf{f'''(1)}}{\mathbf{3!}}(\mathbf{x}-\mathbf{1})^3 + \cdots \quad \text{- - - - - -} \rightarrow (1)$$

Method:18 ⇝ Example:1(Continue)

$$\mathbf{f(x)} = \mathbf{f(1)} + \frac{\mathbf{f'(1)}}{\mathbf{1!}}(\mathbf{x-1}) + \frac{\mathbf{f''(1)}}{\mathbf{2!}}(\mathbf{x-1})^2 + \frac{\mathbf{f'''(1)}}{\mathbf{3!}}(\mathbf{x-1})^3 + \cdots \quad \text{- - - - - -}\blacktriangleright (\mathbf{1})$$

Here, $f(x) = e^x \implies f(1) = e$

$f'(x) = e^x \implies f'(1) = e$

$f''(x) = e^x \implies f''(1) = e$

$f'''(x) = e^x \implies f'''(1) = e$

and so on.

From equation **(1)**,

$$f(x) = e + \frac{e}{1!}(x-1) + \frac{e}{2!}(x-1)^2 + \frac{e}{3!}(x-1)^3 + \cdots$$

$$\therefore e^x = e\left[1 + \frac{1}{1!}(x-1) + \frac{1}{2!}(x-1)^2 + \frac{1}{3!}(x-1)^3 + \cdots\right]$$

Method:18 → 1^{st} Form of Taylor's Series

Example:2 Express $f(x) = 2x^3 + 3x^2 - 8x + 7$ in terms of $(x-2)$.

Solution: Here, $f(x) = 2x^3 + 3x^2 - 8x + 7$

By Taylor's Series,

$$\mathbf{f(x)} = \mathbf{f(a)} + \frac{\mathbf{f'(a)}}{\mathbf{1!}}(\mathbf{x}-\mathbf{a}) + \frac{\mathbf{f''(a)}}{\mathbf{2!}}(\mathbf{x}-\mathbf{a})^2 + \frac{\mathbf{f'''(a)}}{\mathbf{3!}}(\mathbf{x}-\mathbf{a})^3 + \cdots$$

Now, putting $\mathbf{a} = 2$

$$\mathbf{f(x)} = \mathbf{f(2)} + \frac{\mathbf{f'(2)}}{\mathbf{1!}}(\mathbf{x}-\mathbf{2}) + \frac{\mathbf{f''(2)}}{\mathbf{2!}}(\mathbf{x}-\mathbf{2})^2 + \frac{\mathbf{f'''(2)}}{\mathbf{3!}}(\mathbf{x}-\mathbf{2})^3 + \cdots \qquad \mathbf{(1)}$$

Method:18 ⇝ Example:2(Continue)

$$f(x) = f(2) + \frac{f'(2)}{1!}(x-2) + \frac{f''(2)}{2!}(x-2)^2 + \frac{f'''(2)}{3!}(x-2)^3 + \cdots \quad \text{------>} (1)$$

Here, $f(x) = 2x^3 + 3x^2 - 8x + 7 \Rightarrow f(2) = 16 + 12 - 16 + 7 = 19$

$$f'(x) = 6x^2 + 6x - 8 \Rightarrow f'(2) = 24 + 12 - 8 = 28$$

$$f''(x) = 12x + 6 \Rightarrow f''(2) = 24 + 6 = 30$$

$$f'''(x) = 12 \Rightarrow f'''(2) = 12$$

Method:18 ⇝ Example:2(Continue)

From equation (1),

$$f(x) = 19 + \frac{28}{1!}(x-2) + \frac{30}{2!}(x-2)^2 + \frac{12}{3!}(x-2)^3$$

$$f(x) = 19 + \frac{28}{1!}(x-2) + \frac{30}{2!}(x-2)^2 + \frac{12}{3!}(x-2)^3$$

$$f(x) = 19 + 28(x-2) + 15(x-2)^2 + 2(x-2)^3$$

Method:18 → 1^{st} Form of Taylor's Series

Example:3 Find the Taylor's series expansion of $f(x) = \tan x$ in power of $\left(x - \frac{\pi}{4}\right)$ showing at least four non-zero terms.

Solution: Here, $f(x) = \tan x$

By Taylor's Series,

$$\mathbf{f(x)} = \mathbf{f(a)} + \frac{\mathbf{f'(a)}}{\mathbf{1!}}(\mathbf{x} - \mathbf{a}) + \frac{\mathbf{f''(a)}}{\mathbf{2!}}(\mathbf{x} - \mathbf{a})^2 + \frac{\mathbf{f'''(a)}}{\mathbf{3!}}(\mathbf{x} - \mathbf{a})^3 + \cdots$$

Now, putting $\mathbf{a} = \frac{\pi}{4}$

Method:18 ⇝ Example:3(Continue)

Now, putting $\mathbf{a} = \frac{\pi}{4}$

$$f(x) = f\left(\frac{\pi}{4}\right) + \frac{f'\left(\frac{\pi}{4}\right)}{1!}\left(x - \frac{\pi}{4}\right) + \frac{f''\left(\frac{\pi}{4}\right)}{2!}\left(x - \frac{\pi}{4}\right)^2 + \frac{f'''\left(\frac{\pi}{4}\right)}{3!}\left(x - \frac{\pi}{4}\right)^3 + \cdots$$

$\dashrightarrow$ **(1)**

➡ Here, $f(x) = \tan x \Rightarrow f\left(\frac{\pi}{4}\right) = \tan\frac{\pi}{4} = 1$

$$f'(x) = \sec^2 x \Rightarrow f'\left(\frac{\pi}{4}\right) = \sec^2\frac{\pi}{4} = 2$$

$$f''(x) = 2\sec^2 x \cdot \tan x \Rightarrow f''\left(\frac{\pi}{4}\right) = 2\sec^2\frac{\pi}{4}\tan\frac{\pi}{4} = 4$$

$$f''(x) = 2\sec^2 x \cdot \tan x \Rightarrow f''\left(\frac{\pi}{4}\right) = 2\sec^2\frac{\pi}{4}\tan\frac{\pi}{4} = 4$$

$$f'''(x) = 2\sec^2 x + 6\tan^2 x\sec^2 x \Rightarrow f'''\left(\frac{\pi}{4}\right) = 2\sec^2\frac{\pi}{4} + 6\tan^2\frac{\pi}{4}\sec^2\frac{\pi}{4}$$

$= 16$ and so on.

From equation (1),

$$f(x) = 1 + \frac{2}{1!}\left(x-\frac{\pi}{4}\right) + \frac{4}{2!}\left(x-\frac{\pi}{4}\right)^2 + \frac{16}{3!}\left(x-\frac{\pi}{4}\right)^3 + \cdots$$

$$\tan x = 1 + 2\left(x-\frac{\pi}{4}\right) + 2\left(x-\frac{\pi}{4}\right)^2 + \frac{8}{3}\left(x-\frac{\pi}{4}\right)^3 + \cdots$$

Method:19 → 2^{nd} form of Taylor's Series

Method:19 → 2^{nd} Form of Taylor's Series

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots$$

Replace a with x & $x - a$ with h

Taylor's Series: 2^{nd} Form

- If $f(x+h)$ possesses derivatives of all order at a point 'h', then Taylor's series of given function $f(x+h)$ at a point 'h', is given by

$$f(x+h) = f(x) + \frac{h}{1!}f'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \cdots$$

Method:19 → 2^{nd} Form of Taylor's Series

Example:1 Expand $\sin\left(\frac{\pi}{4}+x\right)$ in the power of x. Hence, find the value of $\sin 46°$.

Solution: Let, $f(x+h) = \sin\left(\frac{\pi}{4}+x\right)$

$$\therefore f(x) = \sin x$$

→ By Taylor's Series,

$$\mathbf{f(x+h) = f(x) + \frac{h}{1!}f'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \cdots}$$

Now, putting $\mathbf{x} = \frac{\pi}{4}, \mathbf{h} = x$

Method:19 ⇝ Example:1(Continue)

Now, putting $\mathbf{x} = \frac{\pi}{4}, \mathbf{h} = x$

$$f\left(\frac{\pi}{4} + x\right) = f\left(\frac{\pi}{4}\right) + \frac{x}{1!} f'\left(\frac{\pi}{4}\right) + \frac{x^2}{2!} f''\left(\frac{\pi}{4}\right) + \frac{x^3}{3!} f'''\left(\frac{\pi}{4}\right) + \cdots \quad \text{-------}\rightarrow (1)$$

Here, $f(x) = \sin x \Rightarrow f\left(\frac{\pi}{4}\right) = \sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}$

$$f'(x) = \cos x \Rightarrow f'\left(\frac{\pi}{4}\right) = \cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$f''(x) = -\sin x \Rightarrow f''\left(\frac{\pi}{4}\right) = -\sin\frac{\pi}{4} = -\frac{1}{\sqrt{2}}$$

$$f''(x) = -\sin x \Rightarrow f''\left(\frac{\pi}{4}\right) = -\sin\frac{\pi}{4} = -\frac{1}{\sqrt{2}}$$

$$f'''(x) = -\cos x \Rightarrow f'''\left(\frac{\pi}{4}\right) = -\cos\frac{\pi}{4} = -\frac{1}{\sqrt{2}}$$ and so on.

From equation $(\mathbf{1})$,

$$f\left(\frac{\pi}{4}+x\right) = \frac{1}{\sqrt{2}} + \frac{x}{1!}\left(\frac{1}{\sqrt{2}}\right) + \frac{x^2}{2!}\left(-\frac{1}{\sqrt{2}}\right) + \frac{x^3}{3!}\left(-\frac{1}{\sqrt{2}}\right) + \cdots$$

$$\sin\left(\frac{\pi}{4}+x\right) = \frac{1}{\sqrt{2}}\left[1 + x - \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots\right] \dashrightarrow (\mathbf{2})$$

Method:19 ⇝ Example:1(Continue)

$$\sin\left(\frac{\pi}{4}+x\right) = \frac{1}{\sqrt{2}}\left[1 + x - \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots\right] \dashrightarrow (2)$$

To Find: $\sin 46°$

Now, $\sin 46° = \sin(45° + 1°)$

$$= \sin\left(\frac{\pi}{4} + \frac{\pi}{180}\right) \qquad \left\{\because 1° = \frac{\pi}{180}\right\}$$

$$\approx \sin\left(\frac{\pi}{4} + 0.0174\right)$$

Method:19 ⤳ Example:1(Continue)

$$\approx \sin\left(\frac{\pi}{4} + 0.0174\right)$$

➡ Put $\mathbf{x = 0.0174}$, in equation $\mathbf{(2)}$,

$$\sin\left(\frac{\pi}{4} + x\right) = \frac{1}{\sqrt{2}}\left[1 + x - \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots\right]$$

$$\sin(46°) \approx \frac{1}{\sqrt{2}}\left[1 + (0.0174) - \frac{(0.0174)^2}{2!} - \frac{(0.0174)^3}{3!} + \cdots\right]$$

$$\approx 0.7193$$

Method:19 → 2^{nd} Form of Taylor's Series

Example:2 Use Taylor's series to estimate $\sin 38°$.

Solution: Let, $f(x+h) = \sin(45° - 7°) = \sin 38°$

$$\therefore f(x) = \sin x$$

→ By Taylor's Series,

$$f(x+h) = f(x) + \frac{h}{1!}f'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \cdots$$

Now, putting $\mathbf{x} = \frac{\pi}{4}, h = -7° = -7 \times \frac{\pi}{180} \approx -0.122$

Method:19 ⇝ Example:2(Continue)

Now, putting $\mathbf{x} = \frac{\pi}{4}, h = -7° = -7 \times \frac{\pi}{180} \approx -0.122$

$$f\left(\frac{\pi}{4}+x\right) = f\left(\frac{\pi}{4}\right) + \frac{h}{1!}f'\left(\frac{\pi}{4}\right) + \frac{h^2}{2!}f''\left(\frac{\pi}{4}\right) + \frac{h^3}{3!}f'''\left(\frac{\pi}{4}\right) + \cdots \dashrightarrow (1)$$

➡ Here, $f(x) = \sin x \implies f\left(\frac{\pi}{4}\right) = \sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}$

$$f'(x) = \cos x \implies f'\left(\frac{\pi}{4}\right) = \cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$f''(x) = -\sin x \implies f''\left(\frac{\pi}{4}\right) = -\sin\frac{\pi}{4} = -\frac{1}{\sqrt{2}}$$

Method:19 ⇝ Example:2(Continue)

$$f''(x) = -\sin x \implies f''\left(\frac{\pi}{4}\right) = -\sin\frac{\pi}{4} = -\frac{1}{\sqrt{2}}$$

$$f'''(x) = -\cos x \implies f'''\left(\frac{\pi}{4}\right) = -\cos\frac{\pi}{4} = -\frac{1}{\sqrt{2}}$$ and so on.

From equation $(\mathbf{1})$,

$$f\left(\frac{\pi}{4} + x\right) \approx \frac{1}{\sqrt{2}} + \frac{(-0.122)}{1!}\left(\frac{1}{\sqrt{2}}\right) + \frac{(-0.122)^2}{2!}\left(-\frac{1}{\sqrt{2}}\right) + $$

$$+ \frac{x^3}{3!}\left(-\frac{1}{\sqrt{2}}\right) + \cdots$$

Method:19 ⇝ Example:2(Continue)

$$f\left(\frac{\pi}{4}+x\right) \approx \frac{1}{\sqrt{2}} + \frac{(-0.122)}{1!}\left(\frac{1}{\sqrt{2}}\right) + \frac{(-0.122)^2}{2!}\left(-\frac{1}{\sqrt{2}}\right) +$$

$$+\frac{x^3}{3!}\left(-\frac{1}{\sqrt{2}}\right) + \cdots$$

$\sin 38° = 0.6156$ approximately.

Example:3 State Taylor' series for one variable and hence find $\sqrt{36.12}$.

Write any one form of Taylor's series

Solution: By Taylor's Series,

$$f(x+h) = f(x) + \frac{h}{1!}f'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \cdots \quad \text{- - - - - -} \rightarrow (1)$$

➡ Let, $x = 36,\ h = 0.12$

Then, $f(x+h) = \sqrt{x+h}$,

$$\Rightarrow f(x) = \sqrt{x} \ \Rightarrow f(x) = x^{\frac{1}{2}}$$

Method:19 ⇝ Example:3(Continue)

$$\Rightarrow f(x) = \sqrt{x} \;\; \Rightarrow f(x) = x^{\frac{1}{2}}$$

Here, $f(x) = x^{\frac{1}{2}}$

$$\therefore f'(x) = \frac{1}{2}x^{-\frac{1}{2}}$$

$$\therefore f''(x) = -\frac{1}{4}x^{-\frac{3}{2}}$$

$$\therefore f'''(x) = \frac{3}{8}x^{-\frac{5}{2}}$$ and so on

From equation (1),

$$f(x+h) = f(x) + \frac{h}{1!}f'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \cdots$$

$$\sqrt{x+h} = x^{\frac{1}{2}} + \frac{h}{1!}\left(\frac{1}{2}x^{-\frac{1}{2}}\right) + \frac{h^2}{2!}\left(-\frac{1}{4}x^{-\frac{3}{2}}\right) + \frac{h^3}{3!}\left(\frac{3}{8}x^{-\frac{5}{2}}\right) + \cdots$$

Put, $x = 36,\ h = 0.12$

$$\sqrt{x+h} = (36)^{\frac{1}{2}} + \frac{(0.12)}{1!}\left(\frac{1}{2}(36)^{-\frac{1}{2}}\right) + \frac{(0.12)^2}{2!}\left(-\frac{1}{4}(36)^{-\frac{3}{2}}\right) + \frac{(0.12)^3}{3!}\left(\frac{3}{8}(36)^{-\frac{5}{2}}\right) + \cdots$$

Method:19 ⇝ Example:3(Continue)

$$\sqrt{x+h} = (36)^{\frac{1}{2}} + \frac{(0.12)}{1!}\left(\frac{1}{2}(36)^{-\frac{1}{2}}\right) + \frac{(0.12)^2}{2!}\left(-\frac{1}{4}(36)^{-\frac{3}{2}}\right) + \frac{(0.12)^3}{3!}\left(\frac{3}{8}(36)^{-\frac{5}{2}}\right) + \cdots$$

$$\sqrt{36.12} \approx 6 + 0.01 - 0.000008 + \cdots$$

$$\sqrt{36.12} \approx 6.01$$

Method:20 → Maclaurin's Series

Method:20 → Maclaurin's Series

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots$$

Maclaurin's Series

- If $f(x)$ possesses derivatives of all order at a point '0', then Maclaurin's series of given function $f(x)$ at a point '0', is given by

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$$

Method:20 Maclaurin's Series

Example:1 Express $5 + 4(x-1)^2 - 3(x-1)^3 + (x-1)^4$ in ascending power of x.

Solution: Here, $f(x) = 5 + 4(x-1)^2 - 3(x-1)^3 + (x-1)^4$

By Maclaurin's Series,

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots \dashrightarrow (1)$$

Here, $f(x) = 5 + 4(x-1)^2 - 3(x-1)^3 + (x-1)^4$

$$\Longrightarrow f(0) = 5 + 4 + 3 + 1 = 13$$

Method:20 ⇝ Example:1(Continue)

Here, $f(x) = 5 + 4(x-1)^2 - 3(x-1)^3 + (x-1)^4$

$$\Longrightarrow f(0) = 5 + 4 + 3 + 1 = 13$$

$$f'(x) = 8(x-1) - 9(x-1)^2 + 4(x-1)^3 \Longrightarrow f'(0) = -8 - 9 - 4$$

$$= -21$$

$$f''(x) = 8 - 18(x-1) + 12(x-1)^2 \Longrightarrow f''(0) = 8 + 18 + 12$$

$$= 38$$

$$f'''(x) = -18 + 24(x-1) \Longrightarrow f'''(0) = -18 - 24 = -42$$

$$f^{iv}(x) = 24 \Longrightarrow f^{iv}(0) = 24$$

From equation $(\mathbf{1})$,

$$f(x) = 13 + \frac{(-21)}{1!}x + \frac{38}{2!}x^2 + \frac{(-42)}{3!}x^3 + \frac{24}{4!}x^4$$

$$f(x) = 13 - 21x + 19x^2 - 7x^3 + x^4$$

Example:2 Using Maclaurin's series expand e^x and hence derive expansion of e^{-x}.

Solution: Here, $f(x) = e^x$

By Maclaurin's Series,

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots \dashrightarrow (1)$$

Here, $f(x) = e^x \Longrightarrow f(0) = 1$

$f'(x) = e^x \Longrightarrow f'(0) = 1$

$f''(x) = e^x \Longrightarrow f''(0) = 1$

$f'''(x) = e^x \Longrightarrow f'''(0) = 1$

and so on.

From equation (1),

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

Now, replace x with $-x$

$$e^{-x} = 1 + (-x) + \frac{(-x)^2}{2!} + \frac{(-x)^3}{3!} + \cdots$$

$$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots$$

Example:3 Using Maclaurin's series expand $\sin x$ and $\cos x$.

Solution: Here, $f(x) = \sin x$

By Maclaurin's Series,

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots \quad \text{------} \rightarrow (1)$$

Here, $f(x) = \sin x \Longrightarrow f(0) = 0$

$$f'(x) = \cos x \Longrightarrow f'(0) = 1$$

Method:20 ⇝ Example:3(Continue)

- Here, $f(x) = \sin x \Longrightarrow f(0) = 0$

$$f'(x) = \cos x \Longrightarrow f'(0) = 1$$

$$f''(x) = -\sin x \Longrightarrow f''(0) = 0$$

$$f'''(x) = -\cos x \Longrightarrow f'''(0) = -1$$

$$f^{iv}(x) = \sin x \Longrightarrow f^{iv}(0) = 0$$

$f^{v}(x) = \cos x \Longrightarrow f^{v}(0) = 1$ and so on.

From equation (1),

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$$

Method:20 ⇝ Example:3(Continue)

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$$

$$\sin x = 0 + \frac{1}{1!}x + \frac{0}{2!}x^2 + \frac{(-1)}{3!}x^3 + \frac{0}{4!}x^4 + \frac{1}{5!}x^5 - \cdots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

Expand $\cos x$ by yourself.

$$\cos x = 1 - \frac{x^4}{4!} + \frac{x^6}{6!} - \cdots$$

Method:20 ➡ Maclaurin's Series

Example:4 Expand $e^{x\sin x}$ in power of x up to the terms containing x^6.

Solution: Here, $f(x) = e^{x\sin x}$

➡ We know that,

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

$$x \cdot \sin x = x^2 - \frac{x^4}{3!} + \frac{x^6}{5!} - \frac{x^8}{7!} + \cdots$$

Also, $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$

Method:20 ⇝ Example:4(Continue)

Also, $e^{x} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$

➡ Now, replace x with $x\sin x$

$$e^{x\sin x} = 1 + x\sin x + \frac{(x\sin x)^2}{2!} + \frac{(x\sin x)^3}{3!} + \cdots$$

$$e^{x\sin x} = 1 + \left(x^2 - \frac{x^4}{3!} + \frac{x^6}{5!} - \cdots\right) + \frac{1}{2!}\left(x^2 - \frac{x^4}{3!} + \frac{x^6}{5!} - \cdots\right)^2$$

$$+ \frac{1}{3!}\left(x^2 - \frac{x^4}{3!} + \frac{x^6}{5!} - \cdots\right)^3 + \cdots$$

$$e^{x\sin x}=1+\left(x^2-\frac{x^4}{3!}+\frac{x^6}{5!}-\cdots\right)+\frac{1}{2!}\boxed{\left(x^2-\frac{x^4}{3!}+\frac{x^6}{5!}-\cdots\right)^2}+\boxed{\frac{1}{3!}\left(x^2-\frac{x^4}{3!}+\frac{x^6}{5!}-\cdots\right)^3}+\cdots$$

$$e^{x\sin x}=1+\left(x^2-\frac{x^4}{3!}+\frac{x^6}{5!}-\cdots\right)+\frac{1}{2!}\left[\left\{(x^2)^2+\left(\frac{x^4}{3!}\right)^2+\cdots\right\}+\left\{-2\left(x^2\cdot\frac{x^4}{3!}\right)+\cdots\right\}\right]$$

$$+\frac{1}{3!}\left[\left\{(x^2)^3+\left(\frac{x^4}{3!}\right)^3+\cdots\right\}+\cdots\right]+\cdots$$

$$(a+b+c)^2=a^2+b^2+c^2+2ab+2bc+2ca$$

$$(a+b)^3=a^3+b^3+3ab(a+b)$$

Method:20 ⇝ Example:4(Continue)

$$e^{x\sin x} = 1 + \left(x^2 - \frac{x^4}{3!} + \frac{x^6}{5!} - \cdots\right) + \frac{1}{2!}\left[\left\{(x^2)^2 + \left(\frac{x^4}{3!}\right)^2 + \cdots\right\} + \left\{-2\left(x^2 \cdot \frac{x^4}{3!}\right) + \cdots\right\}\right]$$

$$+ \frac{1}{3!}\left[\left\{(x^2)^3 + \left(\frac{x^4}{3!}\right)^3 + \cdots\right\} + \cdots\right] + \cdots$$

$$e^{x\sin x} = 1 + \left(x^2 - \frac{x^4}{3!} + \frac{x^6}{5!} - \cdots\right) + \left[\left\{\frac{x^4}{2!} + \frac{x^8}{2!\,(3!)^2} + \cdots\right\} - 2\left(\frac{x^6}{2!\,3!}\right) + \cdots\right]$$

$$+ \left[\left\{\frac{x^6}{3!} + \frac{1}{3!}\left(\frac{x^4}{3!}\right)^3 + \cdots\right\} + \cdots\right] + \cdots$$

Method:20 ⇝ Example:4(Continue)

$$e^{x\sin x} = 1 + \left(x^2 \boxed{-\frac{x^4}{3!}} + \boxed{\frac{x^6}{5!}} - \cdots\right) + \left[\left\{\boxed{\frac{x^4}{2!}} + \frac{x^8}{2!\,(3!)^2} + \cdots\right\} - \boxed{2\left(\frac{x^6}{2!\,3!}\right)} + \cdots\right]$$

$$+ \left[\left\{\boxed{\frac{x^6}{3!}} + \frac{1}{3!}\left(\frac{x^4}{3!}\right)^3 + \cdots\right\} + \cdots\right] + \cdots$$

$$e^{x\sin x} = 1 + x^2 + \left(-\frac{1}{3!} + \frac{1}{2!}\right)x^4 + \left(\frac{1}{5!} - \frac{1}{3!} + \frac{1}{3!}\right)x^6 + \cdots$$

$$e^{x\sin x} = 1 + x^2 + \left(-\frac{1}{6} + \frac{1}{2}\right)x^4 + \left(\frac{1}{120}\right)x^6 + \cdots$$

$$e^{x\sin x} = 1 + x^2 + \frac{x^4}{3} + \frac{x^6}{120} + \cdots$$

Remember…

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

$$\sin hx = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \cdots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} - \cdots$$

$$\cos hx = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \cdots$$

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots$$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots$$

Example:5 Expand $\tan^{-1} x$ up to first four terms by Maclaurin's series and hence prove that $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right) = \frac{1}{2}\left(x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots\right)$

Solution: Let, $y = \tan^{-1} x$

$$\frac{dy}{dx} = \frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \cdots$$

Integrating both the sides with respect to x

$$y = \left(x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots\right) + c$$

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots$$

$$y = \left(x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots\right) + c$$

$$\tan^{-1}x = \left(x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots\right) + c \dashrightarrow (1)$$

Put $x = 0$

$$\tan^{-1}0 = (0) + c$$

$$c = 0$$

From equation (1)

$$\tan^{-1}x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots$$

Method:20 ⇝ Example:5(Continue)

To prove: $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)=\frac{1}{2}\left(x-\frac{x^3}{3}+\frac{x^5}{5}-\cdots\right)$

➡ Let $y=\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$

Put $x=\tan\theta$

$$y=\tan^{-1}\left(\frac{\sqrt{1+\tan^2\theta}-1}{\tan\theta}\right)$$

$$=\tan^{-1}\left(\frac{\sqrt{\sec^2\theta}-1}{\tan\theta}\right)$$

Method:20 ⇝ Example:5(Continue)

$$= \tan^{-1}\left(\frac{\sqrt{\sec^2\theta - 1}}{\tan\theta}\right)$$

$$= \tan^{-1}\left(\frac{\sec\theta - 1}{\tan\theta}\right)$$

$$= \tan^{-1}\left(\frac{\frac{1}{\cos\theta} - 1}{\tan\theta}\right)$$

$$= \tan^{-1}\left(\frac{1 - \cos\theta}{\tan\theta \cdot \cos\theta}\right)$$

$$= \tan^{-1}\left(\frac{1 - \cos\theta}{\sin\theta}\right)$$

$$= \tan^{-1}\left(\frac{2\sin^2\left(\frac{\theta}{2}\right)}{2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)}\right)$$

$$= \tan^{-1}\left(\tan\left(\frac{\theta}{2}\right)\right)$$

$$= \tan^{-1}\left(\tan\left(\frac{\theta}{2}\right)\right)$$

$$= \frac{\theta}{2}$$

$$= \frac{1}{2}\tan^{-1}x$$

$$\mathbf{\tan^{-1}x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots}$$

$$= \frac{1}{2}\left(x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots\right)$$

Method:20 → Maclaurin's Series

Example:6 Using Maclaurin's Series expand $(1+x)^{\frac{1}{x}}$ up to the terms x^2.

Solution:

$$(1+x)^{\frac{1}{x}} = e^{\frac{1}{x}\log(1+x)}$$

$$= e^{\frac{1}{x}\left(x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\cdots\right)}$$

$$= e^{\left(1-\frac{x}{2}+\frac{x^2}{3}-\frac{x^3}{4}+\cdots\right)}$$

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$$

$$= e^1 e^{\left(-\frac{x}{2}+\frac{x^2}{3}-\frac{x^3}{4}+\cdots\right)}$$

Method:20 ⇝ Example:6(Continue)

$$= e^1 e^{\left(-\frac{x}{2}+\frac{x^2}{3}-\frac{x^3}{4}+\cdots\right)}$$

$$= e\left[1+\left(-\frac{x}{2}+\frac{x^2}{3}-\frac{x^3}{4}+\cdots\right)+\frac{1}{2!}\left(-\frac{x}{2}+\frac{x^2}{3}-\frac{x^3}{4}+\cdots\right)^2-\cdots\right]$$

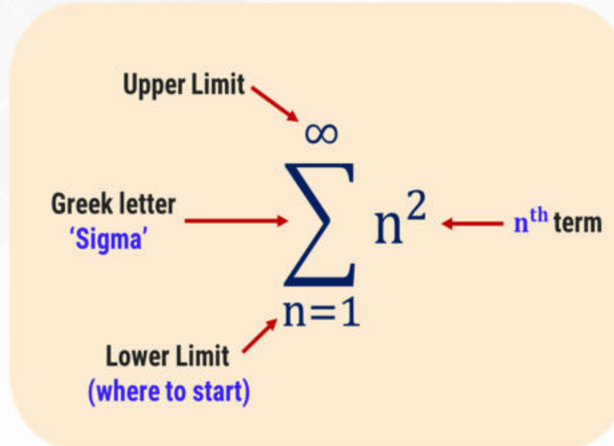
$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$= e\left[1-\frac{x}{2}+\left(\frac{1}{3}+\frac{1}{8}\right)x^2-\cdots\right]$$

$$= e\left[1-\frac{x}{2}+\frac{11}{24}x^2-\cdots\right]$$

Mathematics-I
GTU#(3110014)

Thank You



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